STUDENT TEAM WEEKLY REPORT

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Database Design & Initial Data Preparation]	**TEAM ID:** 2 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 13/08/2026 **STUDENT LEADER:** Akansha Dhage

WORK COMPLETED

- Updated and implemented the PostgreSQL database schema based on the finalized system requirements.
- Added and configured core database tables including User, Role, Subject, Question Bank, Question, Question Option, Quiz, Quiz Attempt, Student Answer, Audit Log, and Subject-Faculty mapping.
- Defined primary keys, foreign keys, constraints, and relationships to maintain referential integrity and support scalable application development.
- Enhanced `Database.sql` with normalized table structures and relationship mappings for user management, question management, quiz generation, and result evaluation.
- Updated `SeedDataWithCredentials.sql` to initialize the database with faculty accounts, subject assignments, student records, default question banks, and audit log entries.
- Integrated collected datasets including Student Data, Faculty Data, MCQ Question Bank, and Descriptive Question Bank into the database initialization workflow.
- Prepared database documentation and schema references for backend integration.

Evidence: Updated `Database.sql`, `SeedDataWithCredentials.sql`, database documentation (`README.md`), faculty and student seed data, question bank datasets.

CHALLENGES / RISKS

- **Amber:** Designing a database structure that supports both MCQ and descriptive questions while maintaining normalization.
- **Amber:** Ensuring correct relationship mapping between users, subjects, question banks, quizzes, and evaluation modules.
- **Green:** Core database schema successfully implemented and validated.
- **Green:** Seed data generation completed using real academic datasets.
- **Green:** Database layer prepared for backend API integration.

HELP NEEDED

- **Technical Guidance Request:** Need review of finalized schema

INTEGRATION AND COLLABORATION

- Collaborated with backend team members to align database entities with planned API endpoints.
- Coordinated with teammates responsible for authentication and quiz modules to ensure compatibility with database structures.
- Verified mapping of faculty, student, and question-bank datasets with the implemented schema.
- Discussed the complete workflow from user management and question storage to quiz attempts, evaluation, and result generation.
- Contributed to preparing the persistence layer required for future backend integration.

COMMON KPI EVIDENCES

- Implemented complete PostgreSQL database schema.
- Created and configured 13+ relational tables with constraints and relationships.
- Developed seed data scripts for students, faculty, subjects, and question banks.
- Established normalized entity relationships and mapping tables.
- Prepared database documentation for future development and integration.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• Perform database testing and validation of schema constraints. • Support backend API integration with database tables. • Implement and verify CRUD operations for major entities.	